MAT 504

Week 3 Hands-on – From Atomic Trajectories to Diffusion

Computer Modelling in Materials Science

Objective. In Week 3, we connected a microscopic model to a macroscopic transport property. Interatomic interactions generate forces; the forces determine accelerations; a numerical integration method generates atomic trajectories; and those trajectories can be analyzed to obtain a diffusion coefficient. In this hands-on exercise, you will use that same framework to answer two practical questions:

1. How does the molecular-dynamics time step affect numerical reliability?
2. How can a trustworthy atomistic trajectory be converted into a diffusion coefficient and then used in a continuum diffusion model?

The central conceptual chain is therefore: the potential-energy model determines forces, the forces are integrated in time to generate trajectories, the trajectories give the mean-squared displacement, the mean-squared displacement gives the diffusion coefficient, and that coefficient can be used in a continuum description.

Computational setup

A starter Jupyter notebook is provided. Complete the indicated `TODO` cells and answer the short questions directly in the notebook. Work through the parts in order; later sections use results generated earlier in the exercise.

The notebook should run from beginning to end without manual intervention. Read the supplied helper functions as well as the code that you complete yourself so that you can connect each computational step to the molecular-dynamics concepts introduced in the lecture.

Model used in this exercise

We use a small two-dimensional Lennard–Jones fluid in reduced units,

$$\epsilon = \sigma = m = k_B = 1,$$

with pair potential

$$U(r) = 4\left(r^{-12} - r^{-6}\right).$$

The particles are placed in a periodic square simulation cell. The supplied simulation engine uses the same **position-Verlet / central-difference idea** introduced in the lecture,

$$\mathbf{r}_i(t + \Delta t) = 2\mathbf{r}_i(t) - \mathbf{r}_i(t - \Delta t) + \mathbf{a}_i(t)\Delta t^2,$$

with

$$\mathbf{a}_i = \frac{\mathbf{F}_i}{m_i}.$$

The notebook stores **unwrapped positions** for trajectory analysis, while periodic images are used internally when evaluating interparticle forces. This allows particles to cross a periodic boundary without artificially resetting their displacement.

The larger time step used below is intentionally chosen to be poor. Do not “fix” it. Its purpose is to show that a trajectory can look plausible while the integration is numerically unreliable.

Part 1 – Predict the effect of the time step

We will compare two simulations that begin from the **same initial configuration and velocities** and run for the same total physical time:

$$\Delta t_{\text{ref}} = 0.005, \qquad \Delta t_{\text{large}} = 0.025.$$

Before running the calculations, answer briefly:

a. Which time step do you expect to conserve the total energy more accurately in an NVE molecular-dynamics trajectory?

b. Why can increasing Δt reduce the accuracy of the Verlet integration even though the same equations of motion are being solved?

Part 2 – Time-step sensitivity and energy conservation

Run the two simulations using the supplied `run_position_verlet` function. Use the parameters already defined in the notebook:

$$N = 36, \qquad \rho = 0.65, \qquad T_0 = 1.0, \qquad t_{\text{total}} = 30.$$

2.1 Relative energy change

For each simulation, calculate

$$\Delta E_{\text{rel}}(t) = 100 \times \frac{E(t) - E(0)}{|E(0)|}.$$

Plot the two curves on the same graph.

2.2 Quantitative comparison

For each run, report:

- the final relative energy change;
- the maximum absolute relative energy deviation over the trajectory.

Present these values in a compact table or clearly formatted output.

2.3 Interpretation

Answer briefly:

a. Which run is numerically more trustworthy?

b. Does the poor run necessarily look obviously "wrong" if you inspect only the particle coordinates?

c. Why is energy conservation a useful diagnostic for an NVE molecular-dynamics calculation?

Part 3 – From atomic trajectories to the diffusion coefficient

Use the **reference-time-step trajectory** for the primary analysis in this section.

3.1 Follow one particle

Choose particle 0 and plot its unwrapped x-y trajectory. Mark its initial and final positions.

In one or two sentences, explain why the path is qualitatively consistent with the microscopic picture of diffusion discussed in the lecture.

3.2 Mean-squared displacement

Calculate the ensemble mean-squared displacement

$$\mathrm{MSD}(t) = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 .$$

Plot MSD versus time for the reference run.

Answer briefly: why is the average over many particles more useful than the displacement of a single particle for estimating a transport property?

3.3 Obtain D

For normal diffusion in d spatial dimensions,

$$\mathrm{MSD}(t) = 2dDt + C.$$

For this two-dimensional system,

$$D = \frac{1}{4}\frac{d\,\mathrm{MSD}}{dt}.$$

Use the supplied fitting interval

$$5 \le t \le 25$$

and fit a straight line to the MSD data. Report the slope and the resulting diffusion coefficient D. Overlay the fitted line on the MSD plot.

3.4 What if the time step is poor?

Repeat only the MSD fit for the large-time-step trajectory and calculate the corresponding apparent diffusion coefficient.

Compare the two values. The two estimates do not have to differ in any particular direction. The important question is whether the large-time-step value should be trusted.

Answer:

> Can a smooth-looking MSD curve or a plausible value of D compensate for poor numerical energy conservation? Why or why not?

Part 4 – Return to the continuum description

Use the diffusion coefficient obtained from the **reference** trajectory in the one-dimensional continuum solution

$$c(x,t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right).$$

Plot $c(x,t)$ for

$$t = 2 \qquad \text{and} \qquad t = 10$$

on the same axes over an appropriate x-range.

Answer briefly:

a. How does the concentration profile change with time?

b. What information from the atomistic simulation is retained in the continuum model through D?

c. What microscopic information is no longer visible once the system is represented only through $c(x,t)$ and D?

Final synthesis

In **3–4 sentences**, explain the complete connection

$$U(\mathbf{r}) \longrightarrow \mathbf{F} \longrightarrow \text{Verlet trajectory} \longrightarrow \text{MSD}(t) \longrightarrow D \longrightarrow c(x,t).$$

Your explanation should make clear which steps are microscopic/atomistic and which step represents the transition to a continuum description.

Before you leave

Make sure your notebook contains:

1. your prediction about time-step accuracy;
2. the two relative-energy-change curves;
3. the numerical energy-conservation comparison;
4. the trajectory of one particle;
5. the reference-run MSD and fitted diffusive region;
6. the diffusion coefficients obtained from the reference and large-time-step runs;
7. the two continuum concentration profiles generated using the reference D;
8. all written answers and the final synthesis.